

Net731 Week 2

20240038

1. Network Design Strategy

Before creating any resources , we need a plan. For **TechBridge Solutions**, we will implement a standard **Two-tier architecture**.

Address Space Choice:

- **Selected Range:** 10.0.0.0/16
- **Justification:** Using the **RFC 1918** private range. A /16 provides 65,536 addresses. While Techbridge is currently small, it allows massive future expansion without needing to re-address the entire network later on in the project.

Subnet Layout

Subnet Name	Address Range	Purpose	Sizing Justification
Public-Subnet	10.0.1.0/24	For web servers/load balancers.	251 usable IPs. Sufficient for a frontend fleet.
Private-Subnet	10.0.2.0/24	For internal databases/backend.	Isolation from the internet for sensitive data.

2. Step-by-Step Configuration in Azure

Task 1: Create the Virtual Network (VNet)

1. Log in to the **Azure Portal**.
2. Search for **Virtual Networks** and click **Create**.
3. **Project Details:** Select your Subscription and create a new Resource Group (RG-TechBridge).
4. **Instance Details:**
 - **Name:** techbridge-Vnet
 - **Region:** (Choose your closest region): North South Africa).
5. **IP Addresses tab:**

- IPv4 Address space: Delete the default and enter **10.0.0.0/16**.
6. Click **Review + create**, then **Create**.

Task 2: Configure Subnets

1. Navigate to your new VNet (techbridge-vnet).
2. Select **Subnets** from the left-hand menu.
3. Click **+ Subnet**.
 - **Name:** Public-Subnet
 - **Address range:** 10.0.1.0/24
 - Click **Save**.
4. Click **+ Subnet** again.
 - **Name:** Private-Subnet
 - **Address range:** 10.0.2.0/24
 - Click **Save**.

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purpose ⓘ Default

Name * ⓘ Private-Subnet

IPv4

Include an IPv4 address space ☒

IPv4 address range ⓘ 10.0.0.0/16
10.0.0.0 - 10.0.255.255

Starting address * ⓘ 10.0.2.0

Size ⓘ /24 (256 addresses)

Subnet address range ⓘ 10.0.2.0 - 10.0.2.255

IPv6

Include an IPv6 address space ☐ This virtual network has no IPv6 address ranges.

Private subnet

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. [Learn more](#)

Add **Cancel** [Give feedback](#)

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purpose

Name *

IPv4

Include an IPv4 address space ☒

IPv4 address range
10.0.0.0 - 10.0.255.255

Starting address *

Size

Subnet address range

IPv6

Include an IPv6 address space ☐ This virtual network has no IPv6 address ranges.

Private subnet

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. [Learn more](#)

[Give feedback](#)

Task 3: Network Security Group (NSG) & Rules

1. Search for **Network security groups** in the portal search bar and click **Create**.
2. **Name:** nsg-techbridge and click **Review + create**.
3. Once created, go to the NSG and select **Inbound security rules**. Add these three rules:

Priority	Name	Port	Protocol	Action	Justification
100	AllowHTTPIn	80	TCP	Allow	Allows public web traffic to reach the frontend.
110	AllowHTTPSIn	443	TCP	Allow	Essential for secure encrypted web traffic.
120	AllowSSH	22	TCP	Allow	(Restricted to your IP) for remote management.

Microsoft...

Upgrade

10

bthengeni0404@gmail....
DEFAULT DIRECTORY (BTHENGE...

Home > Network foundation | Network security groups

NSG-Techbridge

Network security group

Search

MoveDeleteRefreshGive feedback

Custom security rules
3 inbound, 0 outbound

Associated with
2 subnets, 0 network interfaces

Tags [\(edit\)](#)
[Add tags](#)

Filter by name

Port == allProtocol == allSource == all

Destination == allAction == all

Priority ↑↓Name ↑↓Port ↑↓

Inbound Security Rules

100	AllowHTTPIn	80
110	AllowHTTPSIn	443
120	AllowSSH	22
65000	AllowVnetInBound	Any
65001	AllowAzureLoadBalan...	Any
65500	DenyAllInBound	Any

Microsoft Azure

Home > Create NSG-Techbridge
Network security group

Search

OverviewActivity logAccess controlIAMTagsDiagnose and solve problemsResource visualizerSettingsMonitoringAutomationHelp

InboundOutboundNetworkSubnetsPropertiesLocksMonitoringAutomationHelp

Add inbound security rule

NSG-Techbridge

Source ⓘ
Any

Source port ranges * ⓘ
110

Destination ⓘ
Any

Service ⓘ
HTTPS

Destination port ranges ⓘ
443

Protocol
☐ Any
☒ TCP
☐ UDP
☐ ICMPv4
☐ ICMPv6

Action
☒ Allow
☐ Deny

AddCancel

Give feedback

Microsoft Azure

Home > Create > NSG-Techbridge

Add inbound security rule

NSG-Techbridge

Source [ⓘ]

IP Addresses

Source IP addresses/CIDR ranges * [ⓘ]

10.0.1.0 ✓

Source port ranges * [ⓘ]

100 ✓

Destination [ⓘ]

IP Addresses

Destination IP addresses/CIDR ranges * [ⓘ]

10.0.2.0 ✓

Service [ⓘ]

HTTP

Destination port ranges [ⓘ]

80

Protocol

☐ Any

☒ TCP

☐ UDP

☐ ICMPv4

Add **Cancel** [Give feedback](#)

Microsoft Azure

Home > Create > NSG-Techbridge

Add inbound security rule

NSG-Techbridge

Source [ⓘ]

Any

Source port ranges * [ⓘ]

120 ✓

Destination [ⓘ]

Any

Service [ⓘ]

SSH

Destination port ranges [ⓘ]

22

Protocol

☐ Any

☒ TCP

☐ UDP

☐ ICMPv4

☐ ICMPv6

Action

☒ Allow

☐ Deny

Add **Cancel** [Give feedback](#)

- Attach to Subnet:** Select **Subnets** in the NSG menu, click **Associate**, and select techbridge-vnet

README.md Template Snippet:

TechBridge Solutions Cloud Infrastructure

Bongani Thengeni

20240038

Scenario

This project establishes a secure, scalable two-tier network for TechBridge Solutions to separate public web traffic from sensitive backend data.

Address Plan

- VNet: 10.0.0.0/16
- Public Subnet: 10.0.1.0/24
- Private Subnet: 10.0.2.0/24